

BACK PAPER

Elementary Number Theory

Instructor: Ramdin Mawia

Marks: 45

Time: December 28, 2023; 10:00–13:00.

Attempt FIVE problems, selecting at least one from each section. Each question carries 10 marks. The maximum you can score is 45.

COUNTING, PRIMES, DIVISIBILITY, CONGRUENCES

1. Determine all integers n for which $(n-1)! + 1$ is a power of n . **10**
2. Let a and b be coprime positive integers. If $c > ab - a - b$, prove that there are nonnegative integers u and v such that $au + bv = c$. Also show that $ab - a - b$ cannot be expressed as a linear combination of a and b with nonnegative coefficients. **10**

PRIMITIVE ROOTS AND QUADRATIC RESIDUES

3. What do you mean by a *primitive root* for a positive integer n ? Find a primitive root for the prime $p = 53$. **10**
4. State the Reciprocity Law for the Jacobi symbol. Use it (if necessary, together with other reciprocity laws) to evaluate the Jacobi symbol $(2063/2065)$. **10**

ARITHMETIC FUNCTIONS

5. Use Dirichlet's hyperbola method to prove that **10**

$$\sum_{n \leq x} \tau(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x})$$

where $\tau(n)$ denotes the number of positive divisors of $n \in \mathbb{Z}^+$ and γ is the Euler-Mascheroni constant.

6. If f and g are multiplicative arithmetic functions, prove that their Dirichlet convolution $f * g$ is also multiplicative. **10**

BINARY QUADRATIC FORMS, SUMS OF SQUARES

7. Suppose a positive integer n can be expressed as a sum of two relatively prime squares. Prove that every positive divisor of n also has this property. **10**
8. Let f be a positive definite quadratic form. Prove that there exist positive constants C_1 and C_2 (which may depend on the coefficients of f) such that **10**

$$C_1(x^2 + y^2) \leq f(x, y) \leq C_2(x^2 + y^2),$$

for all real numbers x and y .